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The Next Generation OS

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The Next Generation OS

- Very, very early glimpse of the next major OS release after 4.0
  - As a placeholder we'll call it 5.0 for now
- Get prepared, but there's no emergency because we're telling you far in advance
- Today we'll tell you the major things that will affect you
  - Many details aren't yet decided, and can change (including some things said in this talk!)
- Devices expected to ship in 2002 with this OS

The ARM Processor

- Biggest change is to use ARM processors
- There'll be an overlap in which new 68k (i.e., Motorola DragonBall-based) and ARM devices will be shipping
- Plans for 68k support are still under discussion
  - Perhaps full support, perhaps just some of the new features, perhaps just bug fixes

What Does Using ARM Enable?

- Better multimedia (sound and graphic) capabilities
- Decreased power requirements, but more processing power
- Smaller, cheaper, more capable SmartPhones
- More robust security
- Dominant processor architecture, particularly in wireless
- More scalability and differentiation of products

User Scenario

- Users won't have to know what chip they're using; they'll just get the end benefits
  - We don't want to require people to be "techies" just to purchase and install software
  - 68k emulator will let (good) old apps "just work"
- ARM and 68k devices will not have 100% feature parity because 68k can't do everything that ARM can
  - Same situation as now: apps check for OS support of a feature or manager
- Interoperation is intended and expected

Use What You Have Today

- 5.0 will have the same basic API and philosophy
- Use the 3.5 and 4.0 debug ROMs for testing
- Use the Palm OS® Emulator
  - (And fix the things it warns you about!)
- Use the 4.0 APIs
  - Particularly in the mode where it hides structure internals:

```
#define DO_NOT_ALLOW_ACCESS_TO_INTERNALS_OF_STRUCTS
```

Emulation of 68k Applications

- Since the ARM instruction set is different from the 68k set, we will have an emulator so existing, well-behaved applications will work
- Common bugs and hacks can/will prevent successful emulation, so play by the rules!
- This emulator is not POSE: it isn't for debugging, and doesn't emulate the hardware
  - This one is to run 68k code and let it call native OS APIs
- Patching OS traps won't work

Emulator Performance

- Expect very acceptable performance with most emulated applications
  - Nicely comparable to today's devices
  - Actual speed will vary depending on specific ARM chip and speed chosen
- OS routines called by the app are all native and thus will be much faster
  - So the more time in the OS, the better

Emulator Details

- Integers are stored with different “endianness”: bytes are in a different order than on 68k
  - 0x1234 could be 0x12 0x34, or else 0x34 0x12
- The emulator will at times convert the data inside an OS-owned structure to/from the 68k endianness, for when apps must read/write the data directly
  - Can’t and won’t do this for every structure
  - Use the accessor functions added in 4.0
- UI Resources may be converted during installation process

The ARM Processors

- Based on the 4T architecture, like the ARM7 TDMI line
  - Many chips fit this description
  - Thirty silicon manufacturers of just this one
  - No use of Thumb instruction set, for compatibility
- Devices could use ARM9 or XScale (formerly known as StrongARM) etc.
- No need to recompile for the different ARM chips

Developer Tools

- New compilers needed to create native applications on the new chips
- We will provide the compilers, linkers, etc. to drop in to a CodeWarrior or Visual Studio or gcc-style installation
- Will be able to generate ARM-native PRCs, or 68k, or (possibly) “Fat Binaries” with both styles inside
- The Palm OS® Emulator will be enhanced (or cloned) to support ARM emulation

OS Differences

- The Shared library model will be very changed
  - More efficient
  - Allow use of global variables
  - Automatic loading and linking of libraries
- Runtime model differences
  - Mostly only compiler writers will care, except that apps can have globals and C++ vtables any time
- Multithreading for any application, maybe
  - Not generally recommended or suggested, since not everything in the OS is reentrant
  - Not absolutely for sure yet

Other Differences

- Opportunity to remove some historical limits
  - Remove the 64k record/resource limit
  - Allow more than 16 categories (possibly)
  - Globals
  - Etc.
- Wire Formats
  - For beaming and HotSync etc., use a “wire format” to deal with endianness differences
  - Automatic conversion during beaming of known resources

Screens

- More sizes and shapes and resolutions are coming
  - Some even before 4.0
- Possibility of pixel multiplying, so that old applications work just as before
- Possibility of some soft/active Graffiti areas
- APIs will exist so applications can control
- Moral: don’t access screen buffers directly!

Other Topics

- Format of the built-in PIM database records will change to be extensible
  - Compatibility situation TBD (apps and conduits)
  - Library for older devices to use?
- Conduits will need to care about endianness in private record formats
- Security enhancements
  - Some great 3rd-party solutions exist already
  - More added to 4.0
  - Desires for more are being carefully evaluated

What To Do Now and Later

- This talk covers all that we can say for now
  - More details, tools, guidance, etc. will come in 2001
- Code to the 4.0 APIs, using the Debug ROMs and the Palm OS® Emulator
  - Working properly on them is your primary task
  - Play by the API rules
  - Don’t access system structures or circumvent APIs
- Most people don’t yet do these basic things to get compatibility… so now is the time!

To Sum Up

- The next OS, and ARM, are really happening
- Do the usually small (but usually nonzero) amount of work now, so you're ready
  - APIs, POSE and Debug ROMs are your best friends
- Give us feedback!


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